

$\sum_{i=1}^N (1 - \rho)^i \left[\prod_{j=1}^i \zeta_j \right]$. In general, this comparison is ambiguous. In the proof of Proposition 5(b), we show that the latter sum is greater because ζ_j grows rapidly, reflecting the fact that later users presume others before them have fact-checked and do not make independent judgments based on the reliability of the content. In summary, while the provenance policy effectively stops misinformation when identified quickly, it opens a potential gateway for misinformation to be much more widespread. ■

The observation in Example 1 is related to the literature on informational cascades and herding (for example, see Banerjee (1992), Bikhchandani et al. (2021), and Chen and Papanastasiou (2021)). When provenance policies are enacted, agents tend to follow the sharing decisions of those before them instead of making independent inferences about content veracity before sharing. This opens the door for share cascades based on what is sometimes called a “bystander effect”—sharing based on the belief that the content was already vetted by others.

Finally, part (c) of the proposition shows that when $\rho > \rho^*$, the provenance policy reduces misinformation even when the platform responds to this policy by changing the sharing network. This ensures that the platform cannot game the policy by choosing a different network structure that circumvents provenance regulation.

6.2 Censorship

Next, we consider a policy where the platform can censor misinformation that appears on the platform.²² Formally, we model this as the platform being able to adopt a censorship policy that removes $\delta \in (0, 1)$ fraction of the content containing misinformation (with each piece of misinformation removed with probability δ).²³ For simplicity, we assume this content is removed at $t = 0$ before it is even observed by any of the users.

Proposition 6. *There exists $\delta^* < 1$ such that:*

- (a) *If $\delta > \delta^*$, for any fixed sharing network, the censorship policy reduces the virality of misinformation.*
- (b) *If $\delta < \delta^*$, the censorship policy has an ambiguous effect on the virality of misinformation.*
- (c) *If $\delta > \delta^*$, the censorship policy reduces the virality of misinformation even when the platform optimally chooses the sharing network conditional on the policy.*

This result mimics Proposition 5: when censorship can adequately remove most misinformation, it is an effective policy both in given sharing networks (part (a)) and in sharing networks selected by the platform in response to the policy (part (c)). However, when the censorship policy fails to identify some of the misinformation, there is an “implied truth” effect that kicks in for uncensored articles (as empirically identified in Pennycook et al. (2020a)). Bayesian users believe, correctly, that articles are

²²Alternately, it can “tag” the article in question as disputed by outside sources, with analogous implications. See, for example, Facebook’s policies leading up to the 2020 election on labeling suspected misinformation: <https://about.fb.com/news/2019/10/update-on-election-integrity-efforts/>.

²³As in Section 6.1, we think of δ as being a technology parameter related to how effective the regulator is in identifying misinformation. The assumption that the regulator may make type-I errors but not type-II errors (truthful articles are never misidentified, but misinformation is identified with some probability less than one) is adopted for simplicity.